Shang-Jui Ray Kuo

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Research Interests

Fundamental AI research re-examining the inherited architectural choices of modern systems, currently focused on the input interfaces of vision-language models. Hardware and systems background that lets me work close to systems where those defaults originated, while staying on the AI side to propose alternatives.

Education

Stony Brook University — Stony Brook, NY
Ph.D. in Computer Science – Advisor: Prof. Paola Cascante-Bonilla, SPELL Lab — *Aug. 2024 – Present*

National Taiwan University — Taipei, Taiwan
B.S. in Electrical Engineering – GPA: 3.62/4.3 | Last 60 credits: 4.02/4.3 — *Sept. 2019 – June 2023*

Publications

Do VLMs Need Vision Transformers? Evaluating State Space Models as Vision Encoders
- **Shang-Jui Ray Kuo**, Paola Cascante-Bonilla. *Under Review*, 2026. arXiv | Project Page | Code – Featured on Hugging Face Daily Papers; poster, SUNY AI Symposium 2026

Improving Limited Supervised Foot Ulcer Segmentation Using Cross-Domain Augmentation Strategies
- **Shang-Jui Kuo***, Po-Han Huang*, Chia-Ching Lin, Jeng-Lin Li, Ming-Ching Chang. *ICASSP 2024*. IEEE Xplore (* equal contribution)

Research Experience & Projects

Graduate Researcher – SPELL Lab — Sept. 2025 – Present
Stony Brook University | Advisor: Prof. Paola Cascante-Bonilla — *Stony Brook, NY*
- **State space models match Vision Transformers as VLM vision encoders** in a controlled LLaVA-style backbone-swap; SSMs lead on grounding while competitive on VQA at substantially smaller scale
- ImageNet accuracy and naive backbone scaling are unreliable predictors of downstream VLM performance — a result that should reshape how the field selects vision encoders

Research Assistant – COMPAS Lab — Fall 2025
Stony Brook University | Advisors: Prof. Mike Ferdman, Prof. Peter Milder — *Stony Brook, NY*
- **Brought up AMD AI Engine accelerator (VCK5000 FPGA) from scratch**; retargeted an existing PyTorch-to-NPU pipeline via runtime-only modifications — demonstrating the lab's MLISA compiler framework extends to heterogeneous hardware
- Invented an RTL-level on-chip debug methodology under broken-toolchain conditions (no working simulator; 8-hour bitstream cycles); first working matmul kernel on the new accelerator in the lab

Adaptive Chess Tutor — Dual-LoRA TinyLlama Fine-tuning — Spring 2025 + May 2026 repro
- **Programmatic-SFT adapter lifted rule-probe accuracy 2.9% → 94.2%** over base TinyLlama-1.1B; Mixtral-8x7B distillation into a 2nd LoRA on the frozen Phase 1 adapter; runtime α/β blending characterized as a 17-point move-vs-explanation trade-off curve across 47 evaluations (PEFT, QLoRA 4-bit, HF Transformers, SLURM)

Work Experience

AI Researcher & AI Accelerator Engineer — Mar. 2023 – Feb. 2024
Inventec Corporation – AI Center & Digital Center (AI on Chip) — *Taipei, Taiwan*
- **Contributed to commercial NPU IP product** (VectorMesh™) during initial commercialization: RTL image-resize DSP + NPU co-sim/verification environment accelerating model deployment
- **Lightweight CNN** (10–50K params) for stylus-trajectory regression from capacitive sensors; sub-2-pixel accuracy, replacing a hand-tuned rule-based baseline; deployed in panel IC of a top IC design house

Awards & Technical Skills

1st Place + Best Presentation, Inventec Hackathon 2023 (74 teams; AI laptop-antenna sensing) — Oct. 2023

Languages: Python, C++, Verilog, SystemVerilog
ML: PyTorch, Hugging Face (Transformers, PEFT), State Space Models (Mamba family), LoRA / QLoRA, Knowledge distillation, Flash Attention, bf16 mixed-precision, H100/H200 cluster training
Hardware & systems: RTL Design, AMD VCK5000 (AI Engine bring-up), Vitis/Vivado, RISC-V, LLVM IR